COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010-CPE21S3 - Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	7/9/26	**DATE SUBMITTED:**	7/14/26
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ACTIVITY NO. 1.1 **Basic C++ Programming**

1. PROCEDURE OUTPUT

ILO A

```
#include <iostream>
using namespace std;

// Function Prototypes
void findSum(int a, int b);
bool isGreater(int a, int b);
void logicalOperation(bool a, bool b);

int main()
{
    int x, y;
    bool p, q;

    cout << "Enter first number: ";
    cin >> x;

    cout << "Enter second number: ";
    cin >> y;

    findSum(x, y);

    if (isGreater(x, y))
        cout << x << " is greater than " << y << endl;
    else
        cout << x << " is not greater than " << y << endl;

    cout << "\nEnter first boolean value (0 or 1): ";
    cin >> p;

    cout << "Enter second boolean value (0 or 1): ";
    cin >> q;

    logicalOperation(p, q);
```

```
    return 0;
}

// Function Definitions

void findSum(int a, int b)
{
    cout << "\nSum = " << a + b << endl;
}

bool isGreater(int a, int b)
{
    if (a > b)
        return true;
    else
        return false;
}

void logicalOperation(bool a, bool b)
{
    cout << "AND = " << (a && b) << endl;
    cout << "OR = " << (a || b) << endl;
    cout << "NOT A = " << !a << endl;
    cout << "NOT B = " << !b << endl;
}
```

```
Enter first number: 5
Enter second number: 6

Sum = 11
5 is not greater than 6

Enter first boolean value (0 or 1): 0
Enter second boolean value (0 or 1): 1
AND = 0
OR = 1
NOT A = 1
NOT B = 0

--------------------------------
Process exited after 12.44 seconds with return value 0
Press any key to continue . . .
```

ILO B

```
#include <iostream>
using namespace std;

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    bool validateTriangle() const;
};

int main() {
    Triangle set1(40, 30, 110);

    if (set1.validateTriangle()) {
        cout << "The shape is a valid triangle." << endl;
    } else {
        cout << "The shape is NOT a valid triangle." << endl;
    }

    return 0;
}

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

bool Triangle::validateTriangle() const {
    return totalAngle == 180;
}
```

```
The shape is a valid triangle.

--------------------------------
Process exited after 0.01001 seconds with return value 0
Press any key to continue . . .
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C

```
#include <iostream>
using namespace std;

int main() {
    int num1, num2, temp;

    cout << "Enter first number: ";
    cin >> num1;

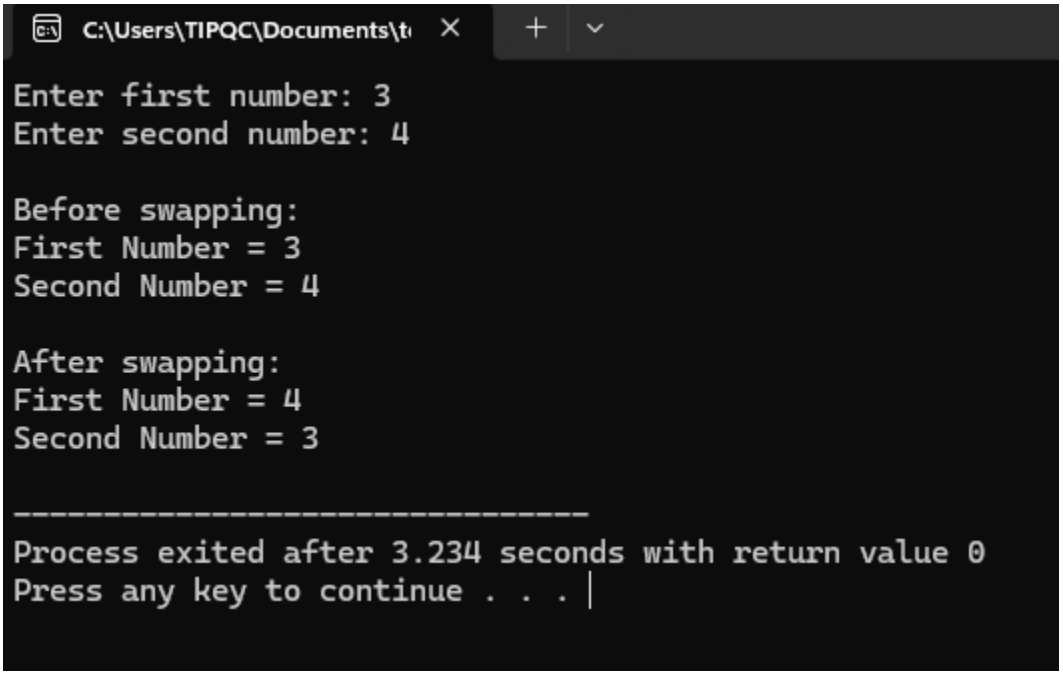
    cout << "Enter second number: ";
    cin >> num2;

    cout << "\nBefore swapping:" << endl;
    cout << "First Number = " << num1 << endl;
    cout << "Second Number = " << num2 << endl;

    temp = num1;
    num1 = num2;
    num2 = temp;

    cout << "\nAfter swapping:" << endl;
    cout << "First Number = " << num1 << endl;
    cout << "Second Number = " << num2 << endl;

    return 0;
}
```




```
#include <iostream>
using namespace std;

double kelvinToFahrenheit(double kelvin) {
    return (kelvin - 273.15) * 9 / 5 + 32;
}

int main() {
    double kelvin;

    cout << "Enter temperature in Kelvin: ";
    cin >> kelvin;

    cout << "Temperature in Fahrenheit: " << kelvinToFahrenheit(kelvin) << endl;

    return 0;
}
```

```
Enter temperature in Kelvin: 243
Temperature in Fahrenheit: -22.27

--------------------------------
Process exited after 5.072 seconds with return value 0
Press any key to continue . . .
```

```
#include <iostream>
#include <cmath>
using namespace std;

double distance(double x1, double y1, double x2, double y2) {
    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
}

int main() {
    double x1, y1, x2, y2;

    cout << "Enter x1 and y1: ";
    cin >> x1 >> y1;

    cout << "Enter x2 and y2: ";
    cin >> x2 >> y2;

    cout << "Distance = " << distance(x1, y1, x2, y2) << endl;

    return 0;
}
```

```
Enter x1 and y1: 32 5
Enter x2 and y2: 2 2
Distance = 30.1496

--------------------------------
Process exited after 9.526 seconds with return value 0
Press any key to continue . . .
```

```
#include <iostream>
#include <cmath>
using namespace std;
```

```
double areaTriangle(double base, double height) {
    return 0.5 * base * height;
}

double perimeterTriangle(double a, double b, double c) {
    return a + b + c;
}

string triangleType(double a, double b, double c) {
    double sides[3] = {a, b, c};

    for (int i = 0; i < 2; i++) {
        for (int j = i + 1; j < 3; j++) {
            if (sides[i] > sides[j]) {
                double temp = sides[i];
                sides[i] = sides[j];
                sides[j] = temp;
            }
        }
    }

    double x = sides[0];
    double y = sides[1];
    double z = sides[2];

    if ((x * x + y * y) > (z * z))
        return "Acute-angled Triangle";
    else if ((x * x + y * y) < (z * z))
        return "Obtuse-angled Triangle";
    else
        return "Others";
}

int main() {
    double base, height;
    double side1, side2, side3;

    cout << "Enter base: ";
    cin >> base;

    cout << "Enter height: ";
```

```
    cin >> height;

    cout << "Enter three sides: ";
    cin >> side1 >> side2 >> side3;

    cout << "\nArea = " << areaTriangle(base, height) << endl;
    cout << "Perimeter = " << perimeterTriangle(side1, side2, side3) << endl;
    cout << "Triangle Type = " << triangleType(side1, side2, side3) << endl;

    return 0;
}
```

```
Enter base: 10
Enter height: 3
Enter three sides: 5
5
5

Area = 15
Perimeter = 15
Triangle Type = Acute-angled Triangle

--------------------------------
Process exited after 10.02 seconds with return value 0
Press any key to continue . . .
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

ChatGPT

Purpose of Use:

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Formatting, debugging and researching

Extent of Use:

Just debugging and searching for possible error causes

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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